

Iteration 1: Falsifiable DGP Estimation for ICHI cKES/USDT on Celo — A Validated Method and an Honest 3/4

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2026-05-29

Abstract

We estimate the data-generating process (DGP) of swap arrivals for the ICHI cKES/USDT LP-aggregator pool on Celo using only free-tier data resources, and evaluate a pre-registered four-criterion falsification gate for a convex (Panoptic-replicated) hedge against a linear alternative. The estimation pipeline works end-to-end and, decisively, *corrected a false null*: an earlier least-squares fit reported no self-excitation, but the canonical maximum-likelihood estimator finds strong self-excitation — branching ratio $\eta \approx 0.600$ (reported as a lower bound), with the likelihood-ratio test rejecting the inhomogeneous-Poisson null ($p = 0.0$) and the held-out Hawkes log-likelihood beating the Poisson baseline by ≈ 114 nats. Three of the four gate criteria pass. The fourth — the held-out time-rescaling Kolmogorov–Smirnov goodness-of-fit test — misses the $\alpha = 0.05$ threshold by a knife-edge (leg-0 $p = 0.0474$ under the locked min-leg aggregator). The verdict is therefore `gate_passes = FALSE` (3/4), with derived `firing_condition = null_strip_unavailable`. We report this honestly: the data *demand* a convex hedge shape (all four convex-dominance conditions pass), but the priced Carr–Madan strip is not calibratable on the degenerate joint distribution at $n \approx 700$. We frame v1.0 as a validated method and an honest near-miss, and name exactly what unlocks certification.

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1 Introduction

The thesis of this project is that a *falsifiable* data-generating-process estimate for a real decentralized-exchange LP-aggregator pool can be produced using only free-tier data resources, and that a **null result is a valid deliverable**. Treating a null as a project failure is itself a discipline violation: a pre-registered gate that does not pass is information, not defeat.

Iteration 1 anchors on the **ICHI** cKES/USDT pool on Celo (contract 0x61Ef8708fc240DC7f9F2c0d81c3124Df2fd8829F). The economic question is whether the swap-arrival process exhibits self-excitation strong enough — together with fat tails, vol-of-vol, and stablecoin-depeg jump risk — to justify a *convex* (Panoptic-replicated) hedge over a *linear* hedge. The unit of account is USD; the data-payment unit is one USDT (the only stablecoin leg in this substrate).

The contribution of v1.0 is twofold. First, a **methodology** that detected and corrected its own false null (§4). Second, an **honest near-miss verdict**: three of the four pre-registered criteria pass on real data, the fourth misses by a knife-edge, and we refuse to narrow the verdict to manufacture a pass (§6).

2 Model

We model the bivariate swap-arrival process as a marked point process and test a **Hawkes** self-exciting specification against an **inhomogeneous Poisson (NHPP)** baseline.

2.1 Bivariate Hawkes intensity

Let $N_j(t)$ count arrivals on leg $j \in \{0, 1\}$ (the two directions of the cKES/USDT pair) over $[0, T]$. The conditional intensity of leg i is

$$\lambda_i(t) = \mu_i + \sum_j \int_0^t \alpha_{ij} e^{-\beta(t-s)} dN_j(s), \quad i \in \{0, 1\}, \quad (1)$$

where $\mu_i > 0$ is the baseline (exogenous) arrival rate, $\alpha_{ij} \geq 0$ is the excitation that a leg- j event exerts on leg- i intensity, and $\beta > 0$ is the common exponential-kernel decay rate. The kernel $\phi(u) = e^{-\beta u}$ is *un-normalized*: its time integral is $\int_0^\infty e^{-\beta u} du = 1/\beta$, so the per-event excitation mass scales as α_{ij}/β (this normalization convention is the crux of the §4 fixture bug). Setting $\alpha_{ij} \equiv 0$ recovers the NHPP baseline, under which the legs are conditionally independent with time-varying rates.

2.2 Branching ratio

The endogeneity of the process is summarized by the **branching ratio**

$$\eta = \rho\left(\frac{\boldsymbol{\alpha}}{\beta}\right), \quad (2)$$

the spectral radius of the normalized adjacency matrix $\boldsymbol{\alpha}/\beta = (\alpha_{ij}/\beta)_{ij}$. Here η is the expected number of direct offspring per event; $\eta \in [0, 1)$ is the stationarity (sub-critical) region, and $\eta = 0$ is the Poisson limit. We enforce $\rho(\boldsymbol{\alpha}/\beta) < 1$ inside the likelihood optimization. Because of the finite-sample downward bias documented in §4, the reported η is a **lower bound** on the true value.

2.3 Boundary-correct likelihood-ratio test

To test H_0 : NHPP ($\eta = 0$) against H_1 : Hawkes ($\eta > 0$), we use the likelihood-ratio statistic

$$\Lambda = 2[\ell_{\text{Hawkes}}(\hat{\theta}_1) - \ell_{\text{NHPP}}(\hat{\theta}_0)]. \quad (3)$$

The Hawkes null parameter $\eta = 0$ sits on the **boundary** of the parameter space, so the standard $\chi^2_{(1)}$ reference is invalid; under the boundary the asymptotic null is the 50:50 mixture $\frac{1}{2}\chi^2_{(0)} + \frac{1}{2}\chi^2_{(1)}$ (Self & Liang, 1987). We do not rely

on the asymptotic form: we calibrate the null distribution of Λ by a parametric bootstrap over 1000 NHPP replicates, scoring both legs on a common time origin $t_0 = 0$ to avoid the absolute-epoch integral inflation that corrupted the earlier estimate (§4).

2.4 Time-rescaling goodness-of-fit (KS)

Under a correctly specified intensity, the **compensator increments** between successive events are unit-rate exponential. Writing the compensator $\Lambda_i(t) = \int_0^t \lambda_i(u) du$, the time-rescaling theorem of Brown et al. (2002) gives, for event times $t_1 < t_2 < \dots$ on leg i ,

$$\Delta_k = \Lambda_i(t_k) - \Lambda_i(t_{k-1}) \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1) \quad \text{under correct specification.} \quad (4)$$

We compare the held-out (last-20%-temporal-block) rescaled increments against $\text{Exp}(1)$ with a one-sample Kolmogorov–Smirnov test per leg and aggregate by the **minimum leg p-value** (a conservative, pre-registered choice). The held-out split guards against in-sample overfit.

2.5 Carr–Madan strip (the target the gate gates)

Conditional on the gate passing, the convex hedge is priced by **static replication** of a payoff $H(\cdot)$ across a strip of out-of-the-money options (Carr & Madan, 1998). For a log-return $x = \log(S_T/S_0)$ with risk-neutral characteristic function $\psi(u)$, the Carr–Madan transform prices a European claim of log-strike $k = \log K$ as

$$C(k) = \frac{e^{-rT}}{\pi} \int_0^\infty \text{Re}[e^{-iuk} \psi(u)] du, \quad (5)$$

with r the risk-free rate and T the maturity. The strip requires a calibrated joint characteristic function ψ for the two legs. When the empirical joint distribution is degenerate (insufficient overlapping sample; §5), ψ is not estimable, the strip integral cannot be formed, and the run terminates in the **null_strip_unavailable** firing condition — *not* in a claim of “no hedge.”

3 Data

The panel is the ICHI cKES/USDT swap stream over Celo block range [67,378,253, 67,896,653] (about six days of Celo proof-of-stake), materialized from cached free-tier sources with content-addressed provenance (**dataHash a72a4ee...**). The raw parquet carries **832 rows**; after the PANEL-04 phantom-transfer filter these reduce to **778 arrival events** (leg-0 = 382, leg-1 = 396). The held-out temporal block holds 83 + 79 events; the training block holds 299 + 317. The five-row Blockscout spot-check (§7) anchors the panel

to on-chain transactions; the reproducibility manifest (§10) pins the panel by sha256.

4 Methodology — the v1.0 contribution

The headline scientific result of Iteration 1 is that the pipeline **corrected a false null**. The verdict on the same real cKES/USDT panel moved from “no self-excitation detectable” to “strong self-excitation, gate 3/4” once three distinct estimator bugs were found and fixed by an independent diagnostic. We present the three runs side by side.

Table 1

run_id	data / estimator	η	verdict	root-cause bug fixed
0afc6af38e24	synthetic, tick normalized- kernel	0.05 (true)	—	(1) mislabeled fixture: label said $\eta = 0.5$, <code>tick.spectral_radius()</code> was 0.05
ae9e3ba17900	real, least-squares fallback	0.0003 (degenerate)	<code>null_lr</code> (artifact)	(2) projection- trick CI + (3) kernel-blind $\beta = 0.1$
bdaf5c7ba5a2	real, scipy canonical free- β AIC	0.600	3/4 , <code>null_strip_unavailable</code>	(corrected unavailable)

The **three root-cause bugs**, each falsified on result-independent grounds by the independent diagnostic:

1. **Mislabeled fixture.** The synthetic validation fixture `synthetic_hawkes_eta_05.parquet` was labeled `expected_branching_ratio = 0.5`, but `tick.SimuHawkesExpKernels` uses the *normalized* kernel $\phi(t) = \alpha\beta e^{-\beta t}$, so tick’s branching ratio is $\rho(\alpha)$, not $\rho(\alpha/\beta)$. The fixture’s actual spectral radius was 0.05 — a near-Poisson panel. The estimator “failing” to recover $\eta = 0.5$ was the estimator *correctly* recovering the $\eta = 0.05$ that was actually simulated.
2. **Projection-trick CI.** The earlier branching-ratio confidence interval [0.283, 0.371] was not a joint-MLE interval: it fit *unconstrained*, then rescaled the adjacency by $\eta_{\text{target}}/\eta_{\text{current}}$ at each grid point and inverted a $\chi^2_{(1)}$ deficit around the *projected* surface. It is a constrained-projection

artifact, now replaced by a genuine `constrained_mle_profile` that re-optimizes the canonical log-likelihood subject to $\rho(\alpha/\beta) = \eta$.

3. **Kernel-blind** $\beta = 0.1$. The production fit was pinned at a decay $\beta = 0.1$ ($1/\beta = 10$ s), far below the data’s ≈ 665 s pooled inter-arrival timescale, so the kernel could not see across events and understated η roughly fourfold. The corrected estimator selects β by AIC over a free grid; the minimum sits at $\beta = 0.001$ ($1/\beta = 1000$ s), a physically sensible, well-separated interior trough.

The corrected estimator is `scipy_canonical_11`: a free- β AIC-selected joint maximum-likelihood fit wrapping the canonical Hawkes log-likelihood, with a common- t_0 likelihood, a stationarity-rejecting objective, and a genuine constrained-MLE confidence interval. Estimator soundness was independently validated: on a *genuine* $\eta = 0.5$ simulation at $n \approx 700$, the estimator recovers $\hat{\eta} \approx 0.54$ (s.d. 0.076), biased *toward zero* — which is why the real-panel estimate is reported as a lower bound.

5 Results

! Verdict

gate_passes = FALSE (3/4). Strong self-excitation ($\eta \approx 0.600$, LR rejects NHPP at $p = 0.0$, held-out Hawkes wins by ≈ 114 nats), but the held-out time-rescaling KS misses $\alpha = 0.05$ by a knife-edge (leg-0 $p = 0.0474$, statistic $D = 0.148$). Derived **firing_condition = null_strip_unavailable**. This is reported as a near-miss / null result; the verdict is NOT a pass and is NOT narrowed.

5.1 Branching ratio and decay selection

The AIC-selected decay is $\beta = 0.001$ ($1/\beta = 1000$ s), with branching ratio $\eta \approx 0.600$ reported as a **lower bound** (about 13% downward finite-sample bias at $n \approx 700$). The decay-AIC table shows a clean, well-separated interior minimum:

Table 2

decay β	$1/\beta$ (s)	AIC
0.0001	10000	∞
0.001	1000	9800.78 (AIC-min)
0.01	100	9833.86
0.1	10	9937.65
1.0	1	10077.41
10.0	0	10117.49

5.2 Likelihood-ratio test and held-out reversal

The bootstrap LR test (1000 NHPP replicates, common $t_0 = 0$, $\alpha = 0.01$) yields an observed statistic of $\Lambda = 561.29$ with $p = 0.0$ — the NHPP null is **rejected**. Out of sample, the Hawkes log-likelihood (-1206.23) beats the NHPP baseline (-1320.63) by ≈ 114 nats — a reversal of the earlier least-squares run, in which NHPP won out of sample.

Table 3

quantity	value
LR observed statistic Λ	561.29
LR p -value	0.0 (rejects NHPP at $\alpha = 0.01$)
bootstrap replicates	1000
held-out ℓ_{Hawkes}	-1206.23
held-out ℓ_{NHPP}	-1320.63
held-out advantage	≈ -114 nats (Hawkes wins)

5.3 The four-criterion gate

Table 4: **gate_passes = FALSE** (3 of 4). The lone failure is the held-out KS criterion.

criterion	value	pass?
LR rejects NHPP ($\alpha = 0.01$)	$\Lambda = 561.3$, $p = 0.0$	PASS
η floor met ($\eta \geq 0.2$)	$\eta = 0.600$	PASS
branching CI excludes zero	lower = 0.001	PASS
held-out KS passes ($\alpha = 0.05$, min-leg)	leg-0 $p = 0.0474$	FAIL

5.4 The KS knife-edge

The held-out time-rescaling KS test is the single criterion that does not pass. The labeling matters and is stated precisely:

- **Leg 0:** p-value $p = 0.0474$ (KS statistic $D = 0.148$, $n = 83$).
- **Leg 1:** p-value $p = 0.0564$ ($n = 79$).
- **Aggregator:** the pre-registered **min-leg** rule takes the smaller leg p-value, $p = 0.0474 < 0.05$, so **ks_held_out_passes = FALSE**.

The miss is 0.0026 below the $\alpha = 0.05$ threshold — a knife-edge. We print the p-value labeled as a p-value: $p = 0.0474$ is *not* the KS statistic (the statistic is

$D = 0.148$). The full unrounded leg-0 p-value is 0.047350333810196134.¹

5.5 Convex-dominance conditions and the firing condition

Separately from the four-criterion *verdict* gate, all **four convex-dominance conditions** pass, indicating that the *shape* of the optimal hedge is convex (fat tails and self-excitation dominate a linear hedge):

Table 5: `any_condition_passed = true` (all four pass). The hedge SHAPE is convexity-justified. The PRICED strip is a separate question (below).

convex-dominance		
condition	evidence	pass?
vol-of-vol > 0	leg-0 vol-of-vol = 1.180	PASS
positive skew + fat tails	skew = 1.849, excess kurt = 3.602	PASS
Hawkes self-excitation ($\eta \geq 0.2$)	$\eta = 0.600$	PASS
USDT-depeg basis jump	literature-range, $N = 64$ LHS, 0 flips	PASS

The data demand convexity, but the precise Carr–Madan strip (Equation 5) is not calibratable: the empirical joint distribution is degenerate at the held-out overlap ($n_{\min} = 79 \leq 101$, the DEPEND-01 lag-radius floor), so the joint characteristic function ψ cannot be estimated and the strip integral cannot be formed. The run therefore derives `firing_condition = null_strip_unavailable` — convexity-justified, calibration-caveated. The detailed evidence block follows.

6 Anti-fishing provenance

Because the KS miss is only 0.0026 from passing, the temptation to narrow the verdict is real. We document, per the halt disposition memo, the **nine post-hoc changes considered and rejected** — none applied:

1. Relaxing the KS α from 0.05 to clear leg-0 $p = 0.0474$ — **not done**.
2. Switching the KS aggregator from min-leg to leg-average / max-leg / single-leg. The leg-average ≈ 0.0519 *would* “pass” — **explicitly not taken**; the min-leg aggregator is locked.
3. Changing the held-out split fraction (0.2) or the temporal-block definition.
4. Dropping or down-weighting the failing leg (leg-0).
5. Switching the residual test (KS $\rightarrow$ Anderson–Darling / Cramér–von Mises / Ljung–Box) to find one that passes.

¹Full precision from `fit_report.json`: leg-0 `p_value` = 0.047350333810196134, `ks_statistic` = 0.14799982742727918; leg-1 `p_value` = 0.05643942824856608. The report uses the single canonical rounding $p = 0.0474$ everywhere.

6. Re-seeding or re-fitting to perturb the held-out KS p across the 0.05 line.
7. Adding “one more robustness” KS variant.
8. Relabeling `gate_passes = FALSE` as any softened or positive-leaning verdict (the disposition memo enumerates four such forbidden relabelings) — **not done**; the verdict reads `FALSE`, full stop.
9. Footnoting the KS miss to present the result as confirmatory.

The refusal to narrow is a **credibility asset**, not a weakness to bury. A 3/4 that survives this list is worth more than a manufactured 4/4.

7 Spot-check

Five panel rows, drawn by a seed deterministically derived from the `run_id` (`int(sha256("bdaf5c7ba5a2"))[:8], 16)`), anchor the panel to on-chain Celo transactions. The seed is recorded so a fresh clone re-derives the identical rows — no cherry-picking. The 832 raw rows reduce to 778 arrival events after the PANEL-04 filter.

Seed: 3812543816 · panel rows: 832 (→ 778 arrival events)

Table 6

row	block	Blockscout tx
347	67595668	0xdb2ff6160e02faaf...
497	67669197	0xc15bda525510f065...
536	67698083	0x14e81d3424ac615c...
656	67793231	0xf54ebbf83b2b8318...
688	67812909	0xceeb4bec51e5275e...

8 Cost-prior sensitivity

The pre-registered sensitivity sweep re-evaluates the four convex-dominance conditions across the locked grid `rate_per_event` $\in \{1, 5, 10\}$ and `USD_per_query` $\in \{2.5 \times 10^{-6}, 5 \times 10^{-6}, 7.5 \times 10^{-6}\}$ — nine cells. The metric is the four **qualitative booleans**, not a dollar margin (no cost-leg model exists, and a dollar- Δ is blocked by the degenerate joint distribution).

All 9/9 cells report `any_condition_passed = true`. `conditions_cost_prior_invariant = true`.

Table 7

<code>rate_per_event</code>	<code>USD_per_query</code>	any condition passed?
1	2.5e-06	true

Table 7

rate_per_event	USD_per_query	any condition passed?
1	5.0e-06	true
1	7.5e-06	true
5	2.5e-06	true
5	5.0e-06	true
5	7.5e-06	true
10	2.5e-06	true
10	5.0e-06	true
10	7.5e-06	true

The conditions are **invariant** to the cost priors across the entire grid. This is the honest finding: the convex-dominance call is **convexity-driven, not cost-prior-driven** — the four conditions derive from the DGP density, not from the cost leg. Caveat: the cost leg is **modeled, not paid** (no x402-on-Celo settlement exists yet per the pre-registration), so the sweep is partly stipulative. Note that cost-prior invariance is *not* itself a pass of the verdict gate (which remains 3/4); the two are kept distinct.

9 Convexity-justified, calibration-caveated — what unlocks certification

The two findings must not be conflated:

- **Convexity-justified.** The hedge *shape* is robust: all four convex-dominance conditions pass, so fat tails and $\eta \approx 0.6$ self-excitation dominate the linear hedge. The data demand convexity.
- **Calibration-caveated.** The *priced* Carr–Madan strip (Equation 5) is not emittable: the joint distribution is degenerate at $n_{\min} = 79 < 101$, so ψ is uncalibratable. This is **null_strip_unavailable, not** “no hedge.”

Two routes unlock certification in a future iteration (v2), each with an honest tradeoff:

1. **Power-law kernel.** Replacing the exponential kernel with a power-law ($\phi(u) \propto u^{-(1+\theta)}$) introduces long memory and likely fixes the held-out KS by capturing the slow-decaying autocorrelation the exponential kernel under-fits. The cost: a power-law tail is *extrapolated* from $n \approx 700$ events, which is thin for tail inference. Better science, but a heavier identification burden at this sample size.
2. **More data.** A longer or denser panel cleans the held-out KS by reducing finite-sample noise and lifts $n_{\min}$ above the joint-distribution floor, making ψ estimable. Cleaner, but slow on free-tier resources.

Both are **named, not executed** — deferred to Iteration 2 / DGP-V2-01.

10 Conclusion and reproducibility

Iteration 1 ships a validated method and an honest 3/4. The pipeline works end-to-end on free-tier data, corrected a false null, and produced a faithful near-miss verdict on real cKES/USDT data: `gate_passes = FALSE` (3/4), `firing_condition = null_strip_unavailable`. The descriptive findings — $\eta \approx 0.600$, LR rejects NHPP, held-out Hawkes reversal, all four convex-dominance conditions — are real and fully reportable; the verdict remains gate-did-not-pass.

Every number in this report is read from the on-disk `bdaf5c7ba5a2/` artifacts; the fit is never recomputed here. Inputs and derived artifacts are pinned by sha256 in [reports/MANIFEST.md](#); run `make verify-reproducibility` to recompute-and-match, and `make report-ichi` to re-render this document.

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